CSS POSITION

What is `position`?

The `position` property controls **how an element is placed on the page** and how `top`, `left`, `right`, `bottom` behave.

Types of Position

1. static (default)
2. relative
3. absolute
4. fixed
5. sticky

Base Structure (Your Code)

```
.container (position: relative, height: 100vh)
  ├── box2
  ├── box3
  ├── box1 (sticky)
  ├── box4
  ├── more boxes...
```

Each `.box` has fixed height and width, margin, and padding.

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1. STATIC

Concept

- Default position for all elements
- Follows normal document flow (top to bottom)
- `top`, `left`, `right`, `bottom` do not work

Example

```
#box1 {
  position: static;
  top: 20px; /* ignored */
}
```

Diagram

```
[Box2]
[Box3]
```

```
[Box1]
[Box4]
```

No movement happens.

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2. RELATIVE

Concept

- Moves relative to its **original position**
- Original space is still reserved

Example

```
#box1 {
  position: relative;
  top: 20px;
  left: 20px;
}
```

What happens

- Moves 20px down and 20px right
- Leaves an empty gap where it originally was

Diagram

```
Before:
[Box1]

After:
(empty space)
      [Box1 shifted]
```

Key idea: visual movement only, layout space remains.

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3. ABSOLUTE

Concept

- Removed from normal flow
- Positioned relative to the **nearest positioned parent**

Important Rule

Parent must have:

```
position: relative | absolute | fixed | sticky
```

Otherwise, it uses the body.

Your Example

```
.container {
  position: relative;
}

#box1 {
  position: absolute;
  bottom: 20px;
  right: 0;
}
```

What happens

- `.container` becomes reference
- `#box1` moves to bottom-right of container

Diagram

```
.container

 -------------------------
|                         |
|   other boxes           |
|                         |
|              [Box1]     |
 -------------------------
```

Key idea: no space reserved, can overlap.

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4. FIXED

Concept

- Positioned relative to the **viewport (screen)**
- Does not move when scrolling

Example

```
#box1 {
  position: fixed;
  top: 20px;
  right: 0;
}
```

Diagram

```
Screen:

           [Box1 fixed]

Scroll down

           [Box1 still here]
```

Key idea: always visible on screen.

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5. STICKY

Concept

- Combination of relative and fixed
- Behaves like relative initially
- Becomes fixed when scroll reaches it

Your Code

```
#box1 {
  position: sticky;
  top: 0px;
}
```

Step-by-Step Behavior

Before scrolling

```
[Box2]
[Box3]
[Box1]
[Box4]
```

Acts like normal (relative behavior)

When scrolling reaches box1

```
[Box1 sticks to top]
[Box4]
[Box5]
```

Diagram

```
Before scroll:
      [Box1]

After scroll:
[Box1 fixed at top]
```

Conditions for Sticky to Work

1. Parent must have height (you used `100vh`, correct)
2. `top`, `left`, etc. must be defined
3. Parent should not have `overflow: hidden`

Important Differences

Relative vs Absolute

Feature	Relative	Absolute
Space reserved	Yes	No
Movement base	itself	parent
Layout flow	stays	removed

Fixed vs Sticky

Feature	Fixed	Sticky
Based on	viewport	scroll position
Scroll behavior	never moves	becomes fixed
Use case	floating elements	sticky headers

Mistake in Your Code

You used duplicate IDs:

```
id="box2" multiple times
```

IDs must be unique.

Correct approach:

```
class="box box2"
```

Final Understanding

Think in this way:

- static: normal flow
- relative: move from original position
- absolute: move inside parent
- fixed: stick to screen
- sticky: scroll and then stick